

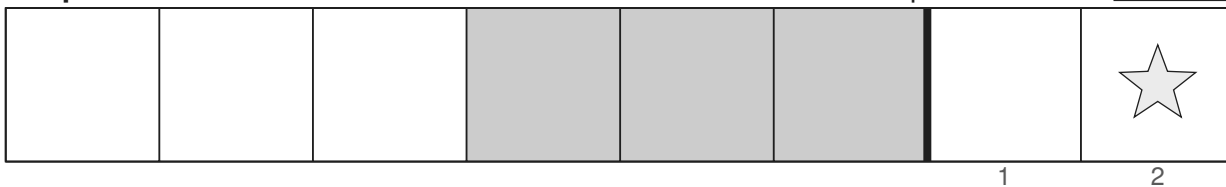
Problem 1: Put a penny on each gray square of strip A. The pennies are the pegs. The thick line is the river. A jump: a peg hops over a peg right next to it, in a straight line, into the empty square just beyond, and the peg it hopped over comes off the strip. Make one jump so that a peg crosses the river. Write an X in the square where it lands.

Strip A

Problem 2: Strips B, C and D start with three, four and six pegs on the home side of the river. For each strip, put pennies on the gray squares, then jump as many times as you like, in either direction, and try to get a peg onto square 2, the square with the star. Write the farthest square any peg reached, 1 or 2, in the blank above the strip. On strip E, choose your own starting squares on the home side, up to six pennies. Draw a dot in each square you used, then play and write the farthest square reached.

Strip B

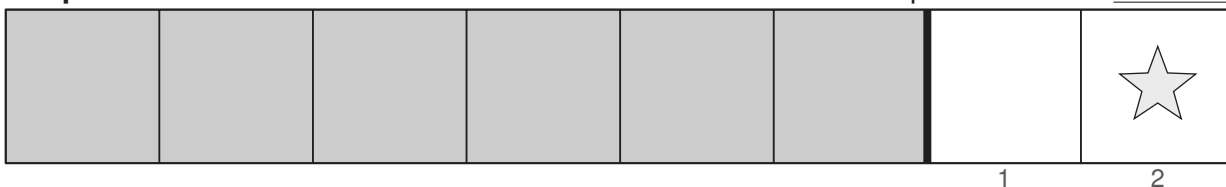
farthest square reached: _____

**Strip C**

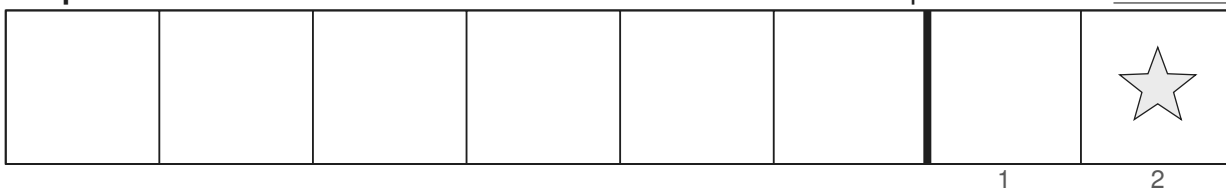
farthest square reached: _____

**Strip D**

farthest square reached: _____

**Strip E**

farthest square reached: _____



Problem 3: Strip F has a value printed in the corner of each square. The total of a position is the sum of the values of the squares that have pegs on them. For each row of the table: put pennies on the squares listed, write the total, make the jump described, and write the new total. In the last column write *up*, *same* or *down*.

Strip F

1	1	2	3	5	8	13	21
							
						1	2

pegs on the squares valued . . .	total before	total after	up, same or down
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all six home squares; make the only jump that is possible			
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2 and 3; jump to the left			
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2 and 3; jump to the right			
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5 and 8; jump to the right, across the river			
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the second 1 and 2; jump to the left			
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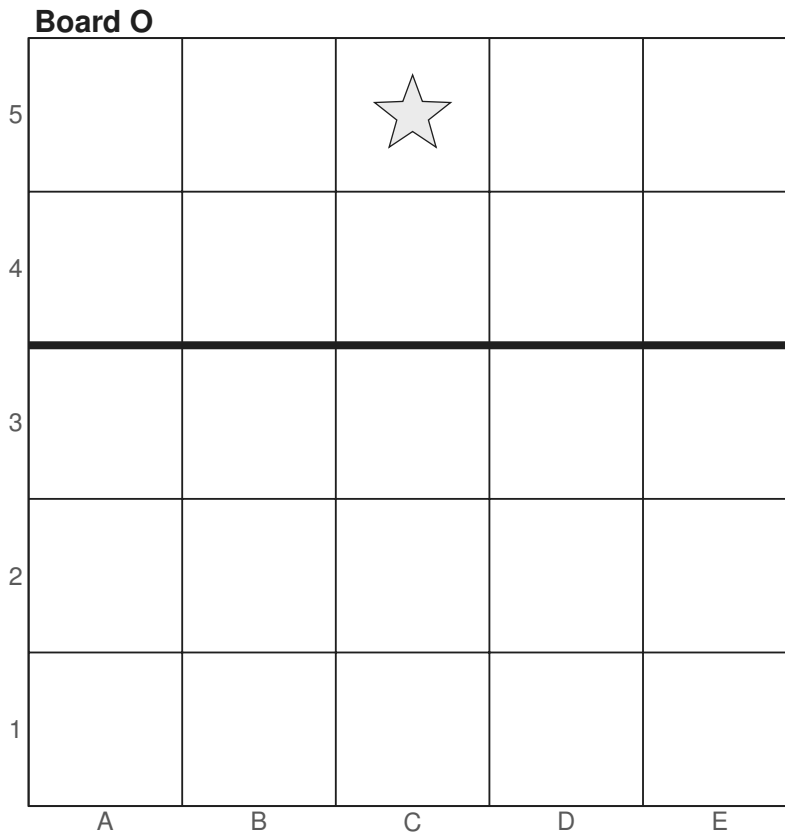
8 and 13, one on each side of the river; jump to the left			
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Problem 4: Write the total of the six home squares of strip F: _____ and the value of square 2: _____. Strip G has seven home squares and no values. Write values in its corners so that the pattern of strip F continues from the left: each value is the sum of the two values before it. Then write the total of the seven home squares: _____ and the value of square 2: _____. Explain to the adult at your table why no peg can ever land on square 2 of strip F, and why the same reason works on strip G.

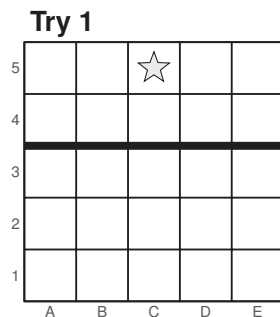
Strip G

							
						1	2

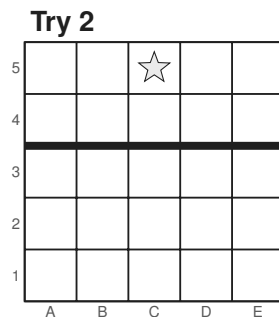
Problem 5: Board O is a grid with the river across it. Squares are named by column letter and row number, like C3. On a board, jumps go up, down, left or right, never diagonally. Put pennies on C3 and C2. Cross the river with one jump. Write the name of the square where the peg lands: _____



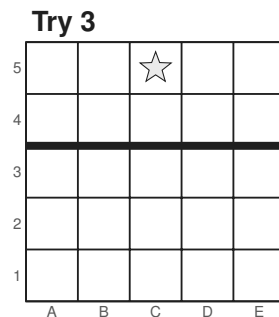
Problem 6: Use board O again. Put only three pennies on home squares of your choice and try to get a peg to the star. Try at least four different placements. Record each placement on one of the small boards below by drawing a dot in each square you used, and write the highest row number any peg reached.



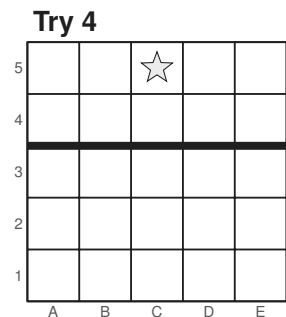
highest row: _____



highest row: _____



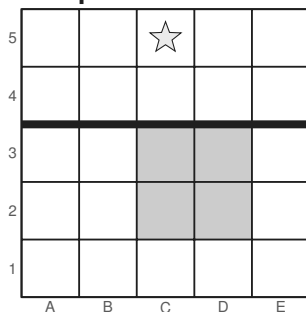
highest row: _____



highest row: _____

Problem 7: Each small picture shows a shape of four pegs on the home side of board O. For each shape, copy it onto board O with pennies and try to get a peg to the star at C5, jumping as many times as you like. Under the picture write *yes* if a peg reached the star and *no* if it never did. For each shape that works, also write its jumps in order, like C1 over C2 to C3.

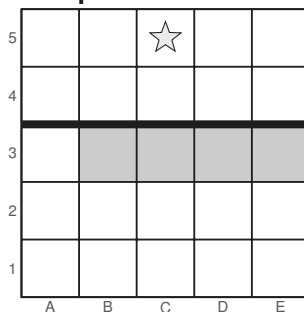
Shape I



works: _____

jumps: _____

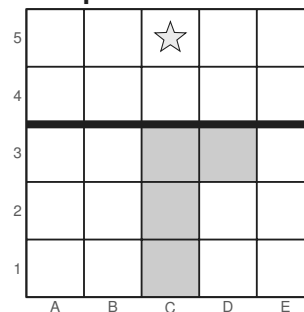
Shape J



works: _____

jumps: _____

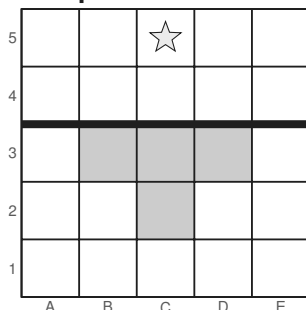
Shape K



works: _____

jumps: _____

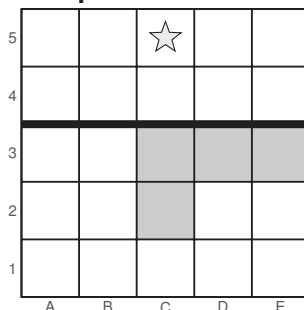
Shape L



works: _____

jumps: _____

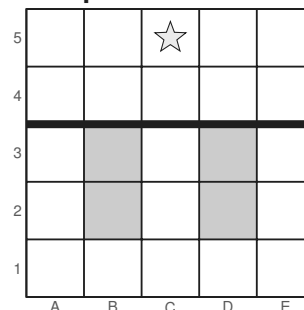
Shape M



works: _____

jumps: _____

Shape N



works: _____

jumps: _____

Problem 8: Board P shows a value in the corner of every square of board O. For each line, put pennies on the squares named, make the jump, and write the total before and the total after.

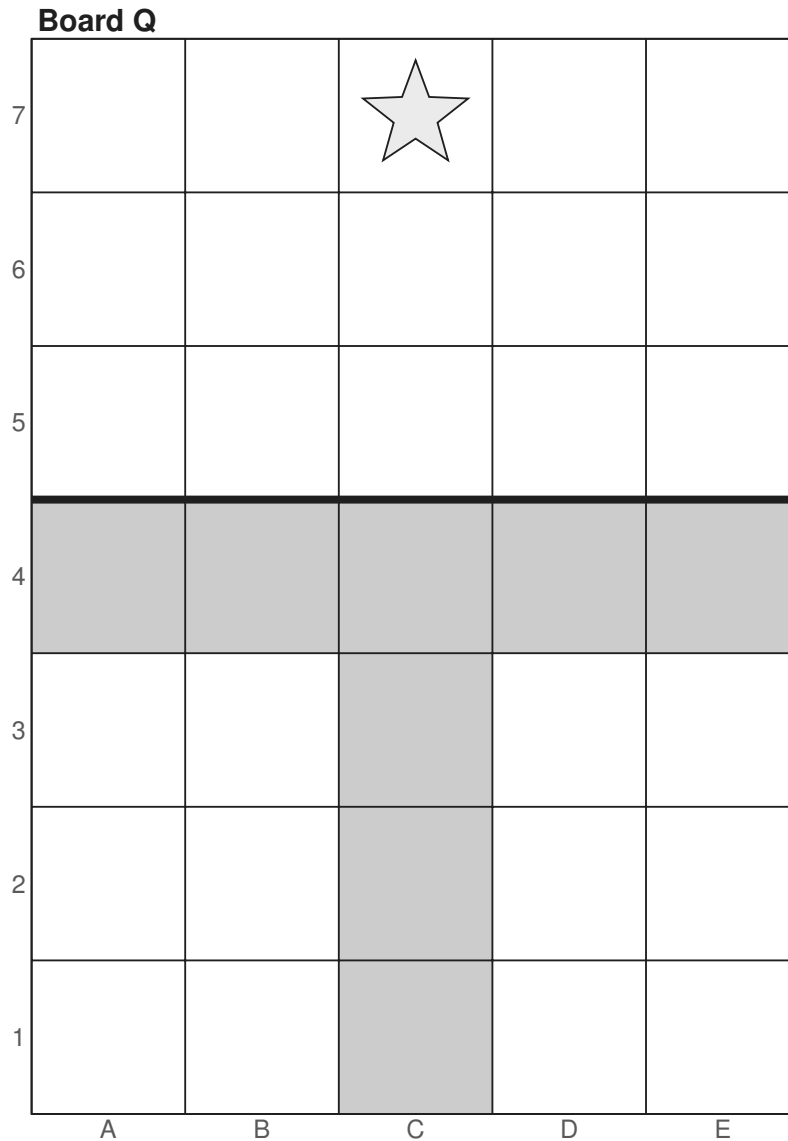
C3 and C2; jump up	before _____	after _____
D3 and E3; jump to the left	before _____	after _____
B3 and C3; jump to the right	before _____	after _____
C4 and C3; jump down	before _____	after _____

Board P

8	13	21	13	8	
5					
5	8	13	8	5	
4					
3	5	8	5	3	
3					
2	3	5	3	2	
2					
1	2	3	2	1	
1					
	A	B	C	D	E

Problem 9: On board P, find the three home squares with the largest values and write their total: _____. Write the value of the star: _____. Write the total of the four squares of shape I: _____ and of shape M: _____. Tell the adult at your table what these totals show about Problem 6, and what shape I shows about whether a total of 21 is enough.

Problem 10: Board Q has eight pegs: five along the river and three below the middle one. Put pennies on the gray squares. Find seven jumps that put a peg on the star at C7. Write down each jump as you make it: the square the peg jumps from, the square it jumps over, and the square it lands on. If you get stuck, set the pegs up again and write a fresh list.



Jump 1: from _____ over _____ to _____

Jump 5: from _____ over _____ to _____

Jump 2: from _____ over _____ to _____

Jump 6: from _____ over _____ to _____

Jump 3: from _____ over _____ to _____

Jump 7: from _____ over _____ to _____

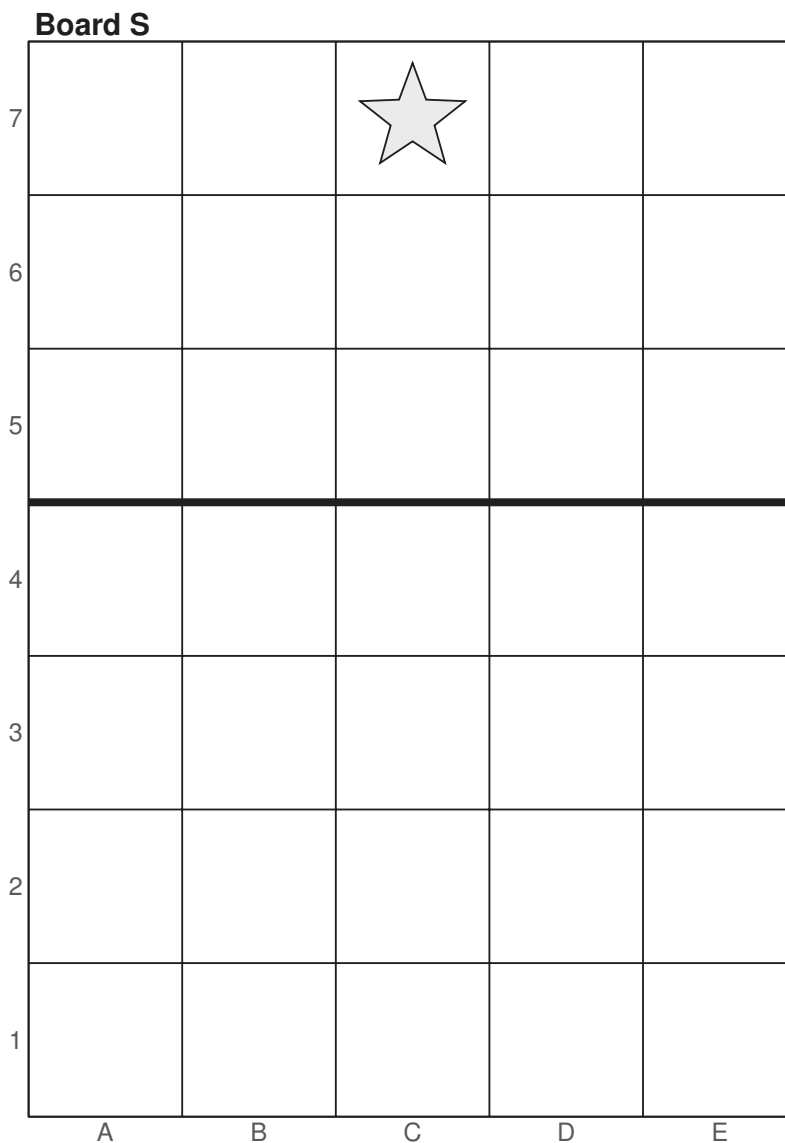
Jump 4: from _____ over _____ to _____

Problem 11: Board R shows the values that go with the star at C7. Write the total of the seven most valuable home squares: _____ and the value of the star: _____. Then write the total of the eight gray squares of board Q: _____. Say what these three numbers tell you about armies of seven pegs and about the army of board Q.

Board R

7	21	34	55 	34	21
6	13	21	34	21	13
5	8	13	21	13	8
4	5	8	13	8	5
3	3	5	8	5	3
2	2	3	5	3	2
1	1	2	3	2	1
	A	B	C	D	E

Problem 12: Board S on the next page is empty. Find a set of eight home squares, different from the shape of board Q, from which a peg can reach the star at C7. Test it with pennies. Draw a dot in each of the eight squares of a set that works, and list its seven jumps.



Jump 1: from _____ over _____ to _____

Jump 5: from _____ over _____ to _____

Jump 2: from _____ over _____ to _____

Jump 6: from _____ over _____ to _____

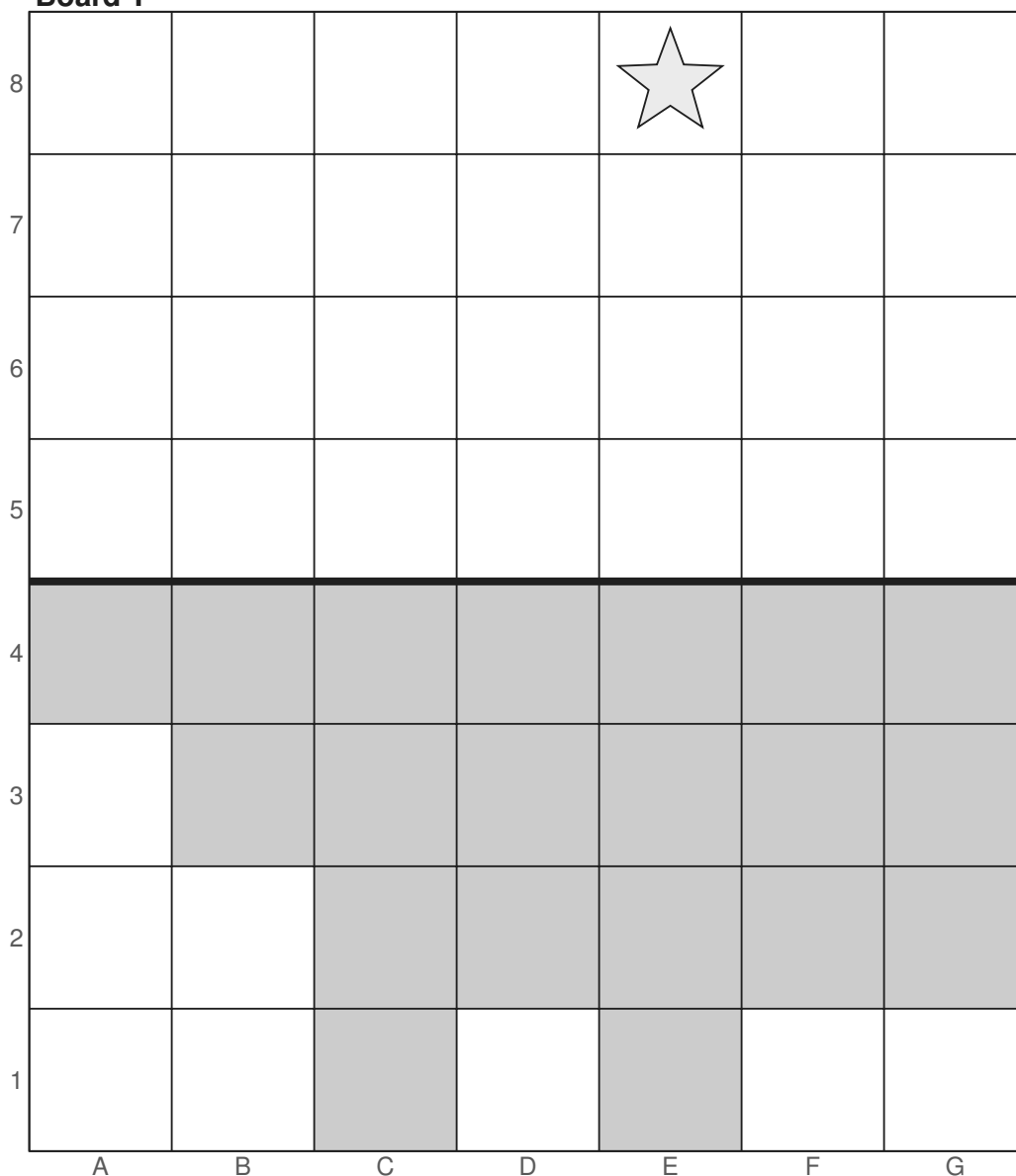
Jump 3: from _____ over _____ to _____

Jump 7: from _____ over _____ to _____

Jump 4: from _____ over _____ to _____

Problem 13: Board T on the next page has twenty pegs, and its star is in the fourth row across the river, at E8. Put pennies on the gray squares and find nineteen jumps that put a peg on the star. Write the jumps down as you make them, on the lines after the board. Before you start, decide which peg should be the last one to move and which two squares must hold pegs just before the final jump.

Board T



Jump 1: from _____ over _____ to _____

Jump 2: from _____ over _____ to _____

Jump 3: from _____ over _____ to _____

Jump 4: from _____ over _____ to _____

Jump 5: from _____ over _____ to _____

Jump 6: from _____ over _____ to _____

Jump 7: from _____ over _____ to _____

Jump 8: from _____ over _____ to _____

Jump 9: from _____ over _____ to _____

Jump 10: from _____ over _____ to _____

Jump 11: from _____ over _____ to _____

Jump 12: from _____ over _____ to _____

Jump 13: from _____ over _____ to _____

Jump 14: from _____ over _____ to _____

Jump 15: from _____ over _____ to _____

Jump 16: from _____ over _____ to _____

Jump 17: from _____ over _____ to _____

Jump 18: from _____ over _____ to _____

Jump 19: from _____ over _____ to _____